

The Optimal-Range Cheat Sheet

The number your lab calls “normal” is built to catch disease — not to tell you you're well. For 18 biomarkers that matter, here's the standard lab range next to the **functional range** associated with actually feeling good and lower long-term risk.

How to read this: pull up your last blood panel and compare each value to both columns. • **Standard lab range** = the reference interval on your report (a disease screen). • **Optimal / functional range** = the tighter band drawn from longevity & outcomes literature. A result can sit comfortably “in range” and still be far from optimal — that gap is usually where how-you-feel lives.

METABOLIC & BLOOD SUGAR

MARKER	STANDARD LAB RANGE	OPTIMAL / FUNCTIONAL	WHY IT MATTERS
Fasting glucose	70–99 mg/dL	75–86 mg/dL	Steady low-normal reflects better insulin sensitivity; the high end of “normal” is drifting toward prediabetes.
HbA1c	< 5.7 %	< 5.4 %	Your ~3-month average glucose. Lower means less glycation and a bigger buffer before prediabetes.
Fasting insulin	2–25 µIU/mL	2–6 µIU/mL	Rises years before glucose does — the earliest sign of insulin resistance. Almost nobody tests it.
HOMA-IR	< 2.9	< 1.5	Combines fasting glucose & insulin into one insulin-resistance score.
Uric acid	3.5–7.2 mg/dL	3.5–5.5 mg/dL	Creeps up with metabolic dysfunction and fructose load well before it causes gout.

HEART & LIPIDS

MARKER	STANDARD LAB RANGE	OPTIMAL / FUNCTIONAL	WHY IT MATTERS
ApoB	< 100 mg/dL	< 80 mg/dL (<70 if higher risk)	Counts the atherogenic particles that actually drive plaque — a better predictor than LDL-C.
Triglycerides	< 150 mg/dL	< 90 mg/dL	High trigs track insulin resistance; the TG/HDL ratio is a quick metabolic read.
HDL	> 40 (M) / > 50 (F)	55–80 mg/dL	Protective, but context matters — very high with high trigs isn't the same story.
Lp(a)	flagged high > 125 nmol/L	< 75 nmol/L (<30 mg/dL)	Genetic and lifelong. Measure it once in your life — it's not on the standard panel.
hs-CRP	< 3.0 mg/L	< 1.0 mg/L	Systemic inflammation. AHA puts low cardiovascular risk under 1.0.
Homocysteine	< 15 µmol/L	< 9 µmol/L	Elevated homocysteine is linked to vascular damage and cognitive decline — and it's cheap to lower.

THYROID & IRON

MARKER	STANDARD LAB RANGE	OPTIMAL / FUNCTIONAL	WHY IT MATTERS
TSH	0.4–4.5 mIU/L	1.0–2.0 mIU/L	Above ~2.5 is linked to fatigue and subtle symptoms even while “in range.”
Ferritin	15–200 (F) / 30–300 (M)	40–100 ng/mL	Low-normal ferritin is one of the most missed causes of fatigue, especially in menstruating women.

VITAMINS

MARKER	STANDARD LAB RANGE	OPTIMAL / FUNCTIONAL	WHY IT MATTERS
Vitamin D (25-OH)	> 30 ng/mL “sufficient”	40–60 ng/mL	The Endocrine Society calls under 30 insufficient; most labs still flag much lower.
Vitamin B12	> 200 pg/mL	> 500 pg/mL	A “normal” total B12 can hide a functional deficiency — check active B12 / MMA if symptomatic.

LIVER & METABOLIC FAT

MARKER	STANDARD LAB RANGE	OPTIMAL / FUNCTIONAL	WHY IT MATTERS
ALT	< 40 U/L	< 25 (M) / < 19 (F)	AASLD lowered the healthy cutoff; a mildly “normal” ALT can flag early fatty liver.
GGT	< 50 U/L	< 25 U/L	Sensitive to fatty liver and oxidative stress; low-normal is a good longevity sign.

Want this done for you? Merios reads every panel against these functional ranges automatically, flags the “in-range-but-suboptimal” gaps, and keeps your trend over time — whatever lab drew it.

Get it on the App Store

Not medical advice. “Optimal / functional” ranges are longevity- and wellness-oriented targets drawn from published literature and organizations including the Endocrine Society, AHA, ADA, AASLD and the EAS; they are *not* diagnostic cutoffs and can vary by age, sex, and clinical context. Standard lab ranges vary by laboratory. Always interpret results with a qualified clinician. © 2026 Merios Health LLC · baseline.merios.life